

Supplementary Materials *for*

**“A Precise and Real-time Eye-Tracking System Powered by Deep
Learning on Smartphones”**

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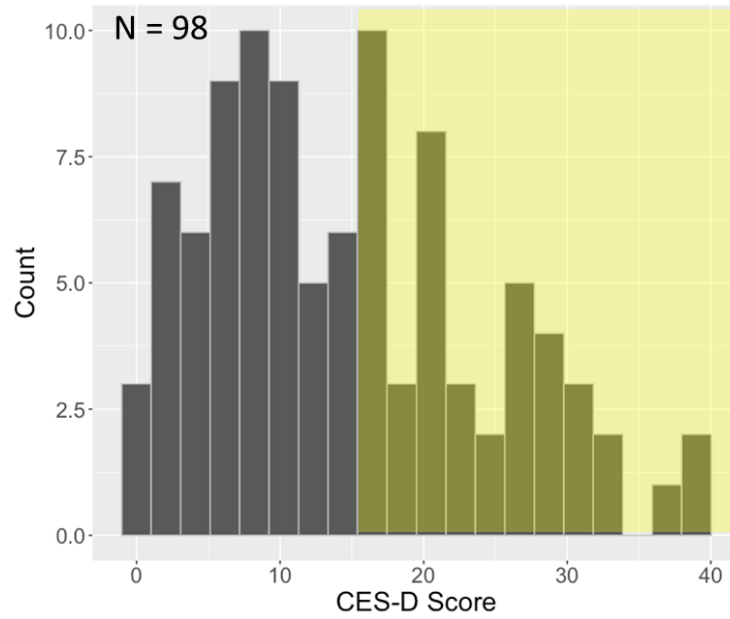


Figure S1. Distribution of the CES-D scores. The yellow area represents samples with CES-D scores equal to or greater than 16.

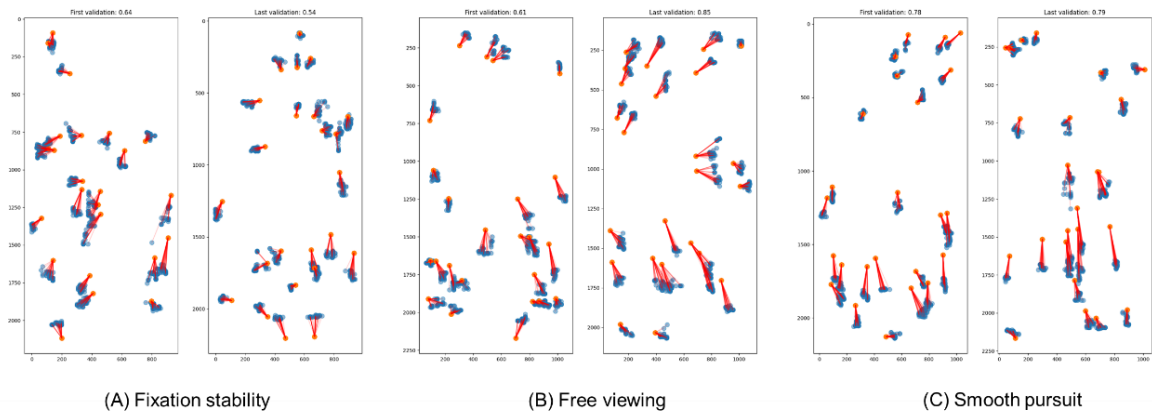


Figure S2. Gaze error in the pre- and post-task evaluation from a representative participant. The red dots represent the ground truth gaze of points, while the blue dots indicate the predicted gaze of points. The red bars represent the gaze error (Euclidean offset).

Table S1. Features extracted from the fixation stability task.

Feature	Description
<i>fs sac count</i>	Saccade counts
<i>fs sacDuration mean</i>	Mean saccade duration
<i>fs sacDuration std</i>	The STD of saccade durations
<i>fs sacAmp mean</i>	Mean saccade amplitude
<i>fs sacAmp std</i>	The STD of saccade amplitudes
<i>fs sacVel mean</i>	Mean saccade velocity
<i>fs sacVel std</i>	The STD of saccade velocities
<i>fs fix count</i>	Fixation count
<i>fs fixDuration mean</i>	Mean fixation durations
<i>fs fixDuration std</i>	The STD of fixation durations
<i>fs x mean</i>	Mean gaze position in the horizontal direction
<i>fs x std</i>	Variation in gaze position in the horizontal direction
<i>fs y mean</i>	Mean gaze position in in the vertical direction
<i>fs y std</i>	Variation in gaze position in the vertical direction

Table S2. Features extracted from the smooth pursuit task.

Feature	Description
<i>sp sac count</i>	Saccade counts
<i>sp sacDuration mean</i>	Mean saccade duration
<i>sp sacDuration std</i>	The STD of saccade durations
<i>sp sacAmp mean</i>	Mean saccade amplitude
<i>sp sacAmp std</i>	The STD of saccade amplitudes
<i>sp sacVel mean</i>	Mean saccade velocity
<i>sp sacVel std</i>	The STD of saccade velocities
<i>sp fix count</i>	Fixation counts
<i>sp fixDuration mean</i>	Mean fixation durations
<i>sp fixDuration std</i>	The STD of fixation durations
<i>sp x offset mean</i>	Mean of gaze offset from the pursuit target in the horizontal
<i>sp x offset std</i>	The STD of gaze offset from the pursuit target in the
<i>sp y offset mean</i>	Mean of gaze offset from the pursuit target in the vertical
<i>sp y offset std</i>	The STD of gaze offset from the pursuit target in the vertical

Table S3. Features extracted from the free viewing task.

Feature	Description
fv_sac_count	Saccade counts
$fv_sacDuration_mean$	Mean saccade duration
$fv_sacDuration_std$	The STD of saccade durations
fv_sacAmp_mean	Mean saccade amplitude
fv_sacAmp_std	The STD of saccade amplitudes
fv_sacVel_mean	Mean saccade velocity
fv_sacVel_std	The STD of saccade velocities
$fv_pic_{\{i\}}_fix_count_top$	Number of fixations (top image)
$fv_pic_{\{i\}}_fixDuration_mean_top$	Mean fixation duration (top image)
$fv_pic_{\{i\}}_fixDuration_std_top$	The STD of fixation durations (top image)
$fv_pic_{\{i\}}_x_mean_top$	The mean gaze position in the horizontal direction (top image)
$fv_pic_{\{i\}}_x_std_top$	The STD of gaze position in the horizontal direction (top image)
$fv_pic_{\{i\}}_y_mean_top$	The mean gaze position in the vertical direction (top image)
$fv_pic_{\{i\}}_y_std_top$	The STD of gaze position in the vertical direction (top image)
$fv_pic_{\{i\}}_fix_count_bot$	Number of fixations (bottom image)
$fv_pic_{\{i\}}_fixDuration_mean_bot$	Mean fixation duration (bottom image)
$fv_pic_{\{i\}}_fixDuration_std_bot$	The STD of fixation durations (bottom image)
$fv_pic_{\{i\}}_x_mean_bot$	The mean gaze position in the horizontal direction (bottom image)
$fv_pic_{\{i\}}_x_std_bot$	The STD of gaze position in the horizontal direction (bottom image)
$fv_pic_{\{i\}}_y_mean_bot$	The mean gaze position in the vertical direction (bottom image)
$fv_pic_{\{i\}}_y_std_bot$	The STD of gaze position in the vertical direction (bottom image)
$fv_pic_{\{i\}}_fix_count_ratio$	$fv_pic_{\{i\}}_fix_count_top / fv_pic_{\{i\}}_fix_count_bot$
$fv_pic_{\{i\}}_fixDuration_mean_ratio$	$fv_pic_{\{i\}}_fixDuration_mean_top / fv_pic_{\{i\}}_fixDuration_mean_bot$
$fv_pic_{\{i\}}_fixDuration_std_ratio$	$fv_pic_{\{i\}}_fixDuration_std_top / fv_pic_{\{i\}}_fixDuration_std_bot$
$fv_pic_{\{i\}}_x_mean_ratio$	$fv_pic_{\{i\}}_x_mean_top / fv_pic_{\{i\}}_x_mean_bot$
$fv_pic_{\{i\}}_x_std_ratio$	$fv_pic_{\{i\}}_x_std_top / fv_pic_{\{i\}}_x_std_bot$
$fv_pic_{\{i\}}_y_mean_ratio$	$fv_pic_{\{i\}}_y_mean_top / fv_pic_{\{i\}}_y_mean_bot$
$fv_pic_{\{i\}}_y_std_ratio$	$fv_pic_{\{i\}}_y_std_top / fv_pic_{\{i\}}_y_std_bot$

Note: $\{i\}$ represents the i -th frame of the free viewing task. Each frame in the free viewing contains two images, presented on the top and bottom areas of the phone screen.

Table S4. Features based on the eye movement data from all three tasks.

Feature	Description
<i>exp_fix_count</i>	Fixation counts
<i>exp_fixDuration_mean</i>	Mean saccade duration
<i>exp_fixDuration_std</i>	The STD of saccade durations
<i>exp_x_mean</i>	The mean gaze position in the horizontal direction
<i>exp_x_std</i>	The STD of gaze position in the horizontal direction
<i>exp_y_mean</i>	The mean gaze position in the vertical direction
<i>exp_y_std</i>	The STD of gaze position in the vertical direction

Table S5. Features with significant differences via the Wilcoxon Rank-Sum Test

Feature name	Non-symptomatic group		Symptomatic group		statistical value	p
	Mean	Std	Mean	Std		
<i>fv_pic_23_y_std_bot</i>	0.5681	0.4946	1.0638	0.7204	(4.0019)	0.0001**
<i>fv_pic_23_y_std_ratio</i>	0.7831	0.1659	0.6548	0.2099	3.1858	0.0014**
<i>fv_pic_4_y_mean_top</i>	6.1563	1.6751	7.0365	1.3548	(3.0390)	0.0024**
<i>fv_pic_34_fixDuration_std_bot</i>	87.7175	53.7123	56.6789	32.8865	3.0140	0.0026**
<i>fv_pic_4_y_mean_ratio</i>	0.3009	0.1452	0.3185	0.1142	(2.8386)	0.0045**
<i>fv_pic_8_y_mean_bot</i>	16.4850	2.8895	17.6497	1.5409	(2.7383)	0.0062**
<i>fv_pic_32_y_std_bot</i>	0.8110	0.6715	1.2080	0.7622	(2.6918)	0.0071**
<i>fv_pic_40_y_std_ratio</i>	0.7210	0.2196	0.5952	0.2404	2.6918	0.0071**
<i>fv_pic_17_x_mean_ratio</i>	0.5484	0.0818	0.5139	0.0449	2.6882	0.0072**
<i>fv_pic_17_fix_count_bot</i>	4.0364	1.6663	4.8837	1.7211	(2.6095)	0.0091**
<i>fv_pic_38_y_std_bot</i>	0.6235	0.5125	0.9073	0.6167	(2.5880)	0.0097**
<i>fv_pic_17_fix_count_ratio</i>	0.5868	0.1522	0.5110	0.1613	2.5701	0.0102*
<i>fv_pic_3_x_mean_ratio</i>	0.5181	0.0981	0.4840	0.0309	2.5164	0.0119*
<i>fv_pic_18_fix_count_bot</i>	4.1818	1.7223	5.0698	1.5946	(2.4878)	0.0129*
<i>fv_pic_27_y_mean_bot</i>	15.9761	3.5929	17.2368	1.5190	(2.4878)	0.0129*
<i>fv_pic_58_x_std_bot</i>	0.7866	0.5588	1.0760	0.7051	(2.4484)	0.0143*
<i>fv_pic_59_y_std_ratio</i>	0.5915	0.2290	0.4821	0.2110	2.3947	0.0166*
<i>fv_pic_35_y_mean_bot</i>	14.3801	4.9160	16.1774	2.9992	(2.3768)	0.0175*
<i>fv_pic_18_fix_count_ratio</i>	0.5769	0.1540	0.5054	0.1390	2.3482	0.0189*
<i>fv_pic_7_y_mean_top</i>	5.1073	1.4729	5.9534	1.5096	(2.3374)	0.0194*
<i>fv_pic_40_y_std_top</i>	2.0625	1.2031	1.4610	1.0022	2.3231	0.0202*
<i>fv_pic_30_fixDuration_std_ratio</i>	0.5993	0.2459	0.4677	0.2672	2.3231	0.0202*
<i>fv_pic_53_fixDuration_std_bot</i>	101.1383	74.2995	73.2006	67.4835	2.3160	0.0206*
<i>fv_pic_53_x_std_ratio</i>	0.7068	0.1838	0.7924	0.1848	(2.3160)	0.0206*
<i>fv_sacVel_mean</i>	0.0446	0.0067	0.0475	0.0064	(2.3088)	0.0210*
<i>fv_sacVel_std</i>	0.0478	0.0078	0.0510	0.0069	(2.3088)	0.0210*
<i>fv_pic_6_y_mean_bot</i>	17.5033	2.9148	18.5586	1.3856	(2.3016)	0.0214*
<i>fv_pic_45_x_std_ratio</i>	0.6049	0.2597	0.4977	0.2235	2.2265	0.0260*
<i>fv_pic_12_y_mean_bot</i>	16.9604	3.6179	18.1693	1.5369	(2.2014)	0.0277*
<i>fv_pic_33_y_std_ratio</i>	0.7194	0.2035	0.6102	0.2487	2.1943	0.0282*
<i>fv_pic_45_y_mean_bot</i>	16.3232	3.7740	17.1309	3.0722	(2.1799)	0.0293*
<i>fv_pic_20_fixDuration_mean_ratio</i>	0.4828	0.1222	0.5268	0.1131	(2.1764)	0.0295*
<i>fv_pic_34_x_mean_ratio</i>	0.4979	0.1044	0.4747	0.0924	2.1513	0.0315*
<i>fv_pic_44_y_std_ratio</i>	0.6881	0.2424	0.6045	0.2189	2.1477	0.0317*
<i>fv_pic_23_x_mean_ratio</i>	0.5453	0.1174	0.5058	0.0458	2.1084	0.0350*
<i>fv_pic_30_fixDuration_std_top</i>	160.7702	129.5734	114.6883	81.7675	2.1012	0.0356*
<i>fv_pic_58_y_std_ratio</i>	0.6071	0.2783	0.5012	0.2291	2.0976	0.0359*
<i>fv_pic_31_x_std_bot</i>	0.4946	0.3409	0.6607	0.3954	(2.0797)	0.0376*
<i>fv_pic_31_y_std_bot</i>	0.9991	0.7601	1.3484	0.8402	(2.0797)	0.0376*
<i>fv_pic_51_x_std_top</i>	0.5961	0.3462	0.8246	0.6067	(2.0797)	0.0376*

<i>fv_pic_39_y_std_ratio</i>	0.6501	0.2473	0.5497	0.2354	2.0797	0.0376*
<i>fv_pic_9_y_mean_bot</i>	16.0064	3.5515	16.9652	3.2616	(2.0726)	0.0382*
<i>fv_pic_58_fix_count_bot</i>	4.1818	2.2202	5.0698	1.8438	(2.0726)	0.0382*
<i>fv_pic_53_x_std_bot</i>	0.6582	0.5675	0.4675	0.5656	2.0654	0.0389*
<i>fv_pic_32_y_std_ratio</i>	0.7011	0.2086	0.6205	0.1841	2.0618	0.0392*
<i>fv_pic_18_y_std_bot</i>	1.0850	0.8697	1.4110	0.8982	(2.0511)	0.0403*
<i>fv_pic_19_fixDuration_std_bot</i>	71.9997	50.0821	84.9690	49.4894	(2.0368)	0.0417*
<i>fv_pic_24_y_std_bot</i>	0.9933	0.9044	1.1947	0.6598	(2.0368)	0.0417*
<i>fv_pic_36_y_std_bot</i>	0.7682	0.5706	1.0643	0.7274	(2.0224)	0.0431*
<i>fv_pic_23_fixDuration_std_bot</i>	72.9422	47.6877	88.4416	41.1281	(2.0189)	0.0435*
<i>fv_pic_48_y_std_top</i>	1.8708	1.0903	1.4614	0.8808	2.0153	0.0439*
<i>fv_pic_43_fixDuration_std_ratio</i>	0.5025	0.2512	0.5878	0.1969	(2.0153)	0.0439*
<i>fv_pic_42_y_std_bot</i>	1.0711	0.6592	1.3732	0.7554	(1.9974)	0.0458*
<i>fv_pic_21_x_mean_ratio</i>	0.5180	0.0765	0.5020	0.0894	1.9902	0.0466*
<i>fv_pic_23_fixDuration_std_ratio</i>	0.6142	0.2206	0.5399	0.1651	1.9831	0.0474*
<i>fv_pic_12_x_mean_ratio</i>	0.5646	0.0928	0.5342	0.0414	1.9795	0.0478*
<i>fv_pic_33_x_mean_bot</i>	5.2737	1.9617	5.9518	1.3331	(1.9759)	0.0482*
<i>fv_pic_3_y_mean_bot</i>	15.3580	3.4271	16.4958	1.5081	(1.9723)	0.0486*
<i>fv_pic_32_x_std_bot</i>	0.8539	0.6710	1.0828	0.5930	(1.9687)	0.0490*

Note: * $p < .05$. ** $p < .001$.